

COMPUTER NETWORKS



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OUTLINE

- Course Objective
- Evaluation Scheme and Books
- Network fundamentals
- Types of networks
- Protocols and standards
- Packet switching and circuit switching
- Significance of layered network architectures
- Transmission Media or IEEE 802.3 Ethernet cables
- Different Types of Delays in the Network
- Reference models

COURSE OBJECTIVE

- To understand the computer network models and their architecture.
- To understand the protocols for each layers in the OSI and TCP/IP model.
- To learn about the MAC layer issues and routing protocols
- To lean transport layer and application layer protocols.

EVALUATION SCHEMA

Mid Semester Exam :	30 (18 Hrs.)
End Semester Exam :	50 (22 Hrs.)
Teacher Assessment : (Assignments/Quiz/Viva)	20

BOOK(S)

- James F. Kurose, Keith W. Ross, “Computer Networking: A Top-Down Approach Featuring the Internet”, 3rd Edition , Pearson Education 2009.
- William Stallings, “Data and Computer Communications”, Seventh Edition, PHI 2004.
- Andrew S. Tanenbaum, ”Computer Networks” 4th Edition PHI
- B. A. Fourozan, “TCP/IP Protocol Suite”, 3rd Edition, Singapore, McGrawHill, 2004.
- L. L. Peterson and B. S. Davie, Computer Networks-A System Approach, Elsevier.
- B. A. Fourozan, “Data Communications and Networking”, 4th Edition, Singapore, McGrawHill, 2004.

COMPUTER NETWORKS

- A computer network is a set of computers sharing resources located on or provided by network nodes.
- The computers use common communication protocols over digital interconnections to communicate with each other.
- These interconnections are made up of telecommunication network technologies, based on physically wired, optical, and wireless radio-frequency methods that may be arranged in a variety of network topologies.
 - Source: wikipedia.org

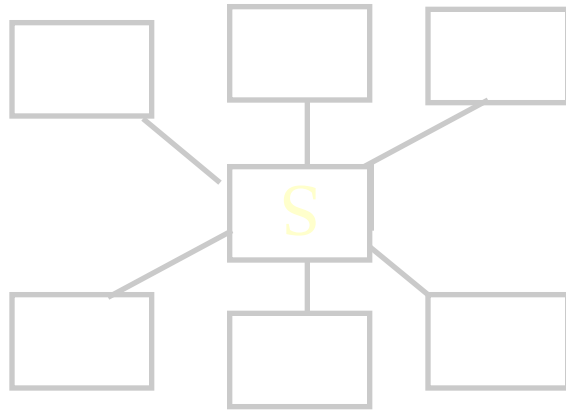
A COMMUNICATIONS MODEL

- Source
 - generates data to be transmitted
- Transmitter
 - Converts data into transmittable signals
- Transmission System
 - Carries data
- Receiver
 - Converts received signal into data
- Destination
 - Takes incoming data

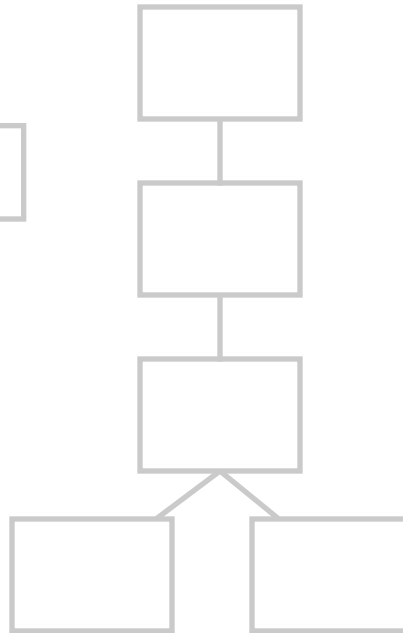
NETWORKS: WHY?

- Potential of networking:
 - move bits *everywhere, cheaply*, and with desired *performance characteristics*
 - *Break the space barrier for information*
- Network provides “connectivity”
 - Direct or indirect access to every other node in the network
 - Connectivity is the magic needed to communicate if you do not have a direct pt-pt physical link.

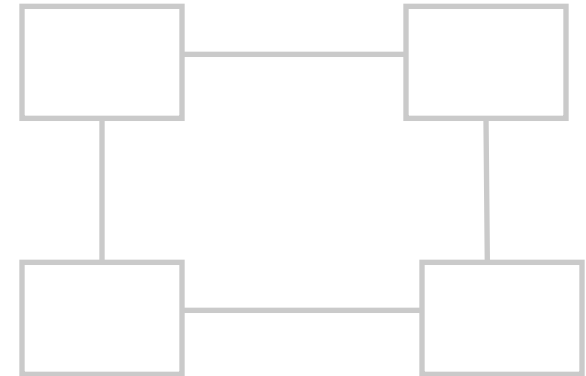
TOPOLOGIES: INDIRECT CONNECTIVITY



Star



Tree



Ring

NETWORK TOPOLOGY:

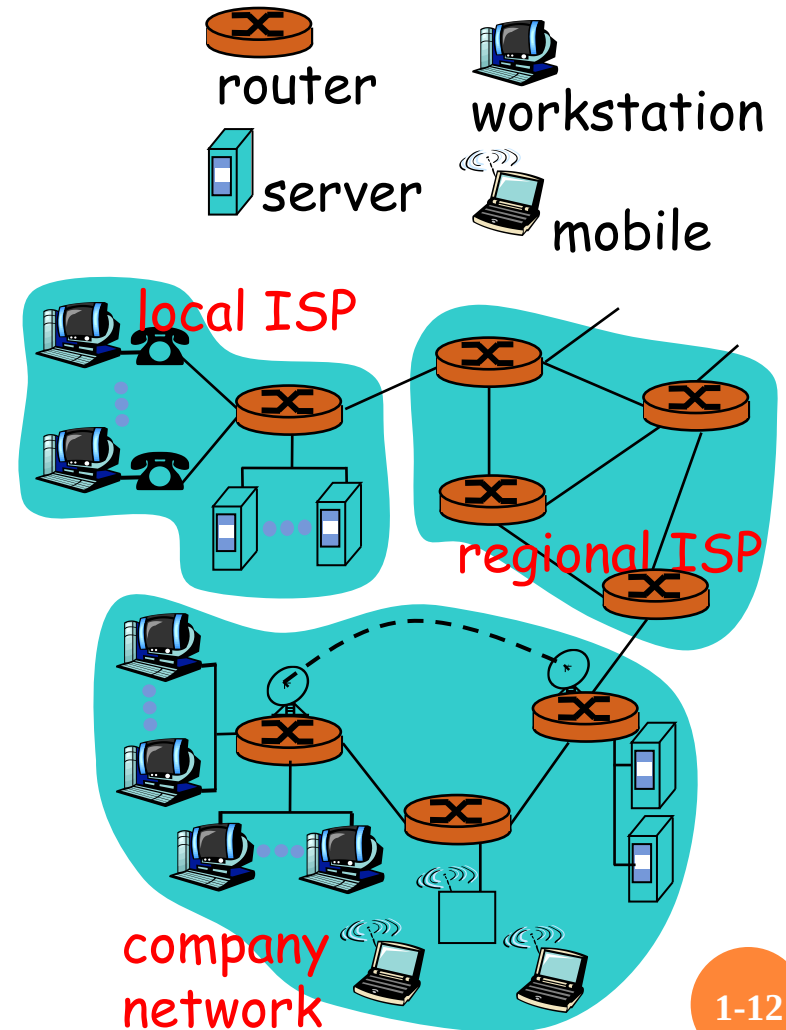
- Bus topology
- Star topology
- Ring topology
- Tree topology
- Mesh topology (Fully Connected)
- Partially Connected Mesh Topology
- Reference
 - <https://www.geeksforgeeks.org/types-of-network-topology/>
 - <https://www.javatpoint.com/computer-network-topologies>

TYPES OF NETWORKS / NETWORKS CLASSIFICATION

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

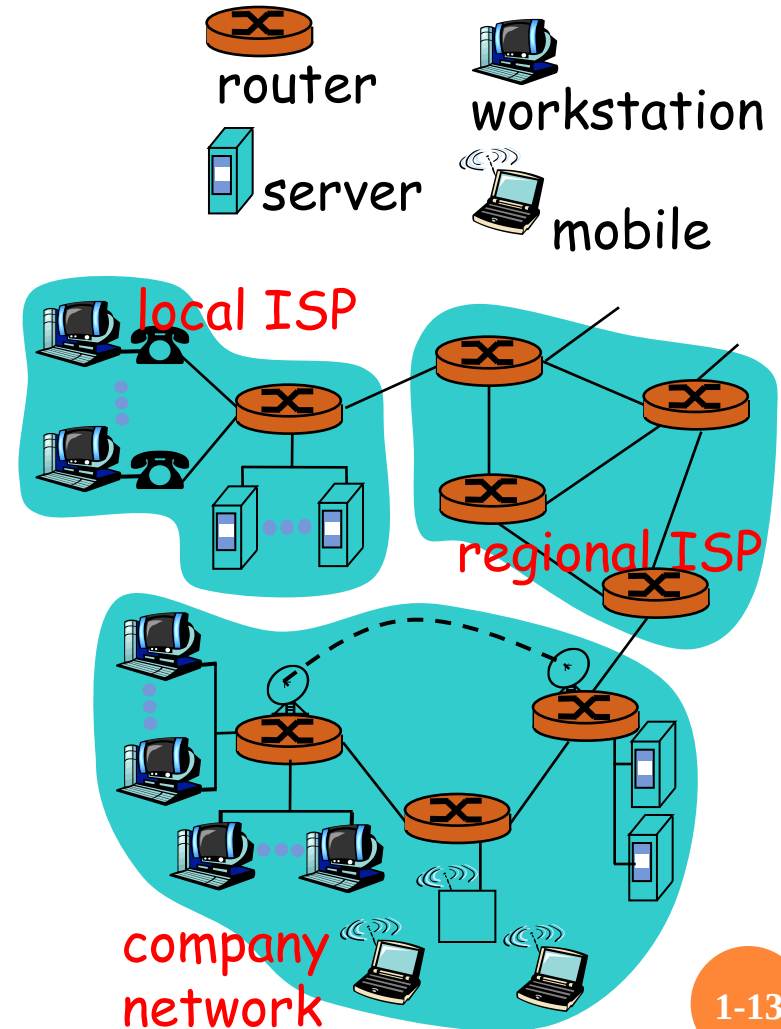
WHAT'S THE INTERNET

- millions of connected computing devices: *hosts = end systems*
- running *network apps*
- *communication links*
 - fiber, copper, radio, satellite
 - transmission rate = *bandwidth*
- *routers*: forward packets (chunks of data)



WHAT'S THE INTERNET

- *protocols* control sending, receiving of msgs
 - e.g., TCP, IP, HTTP, FTP, PPP
- *Internet*: “network of networks”
 - loosely hierarchical
 - public Internet versus private intranet
- Internet standards
 - RFC: Request for comments
 - IETF: Internet Engineering Task Force



WHAT'S A PROTOCOL?

human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific msgs sent

... specific actions taken
when msgs received,
or other events

network protocols:

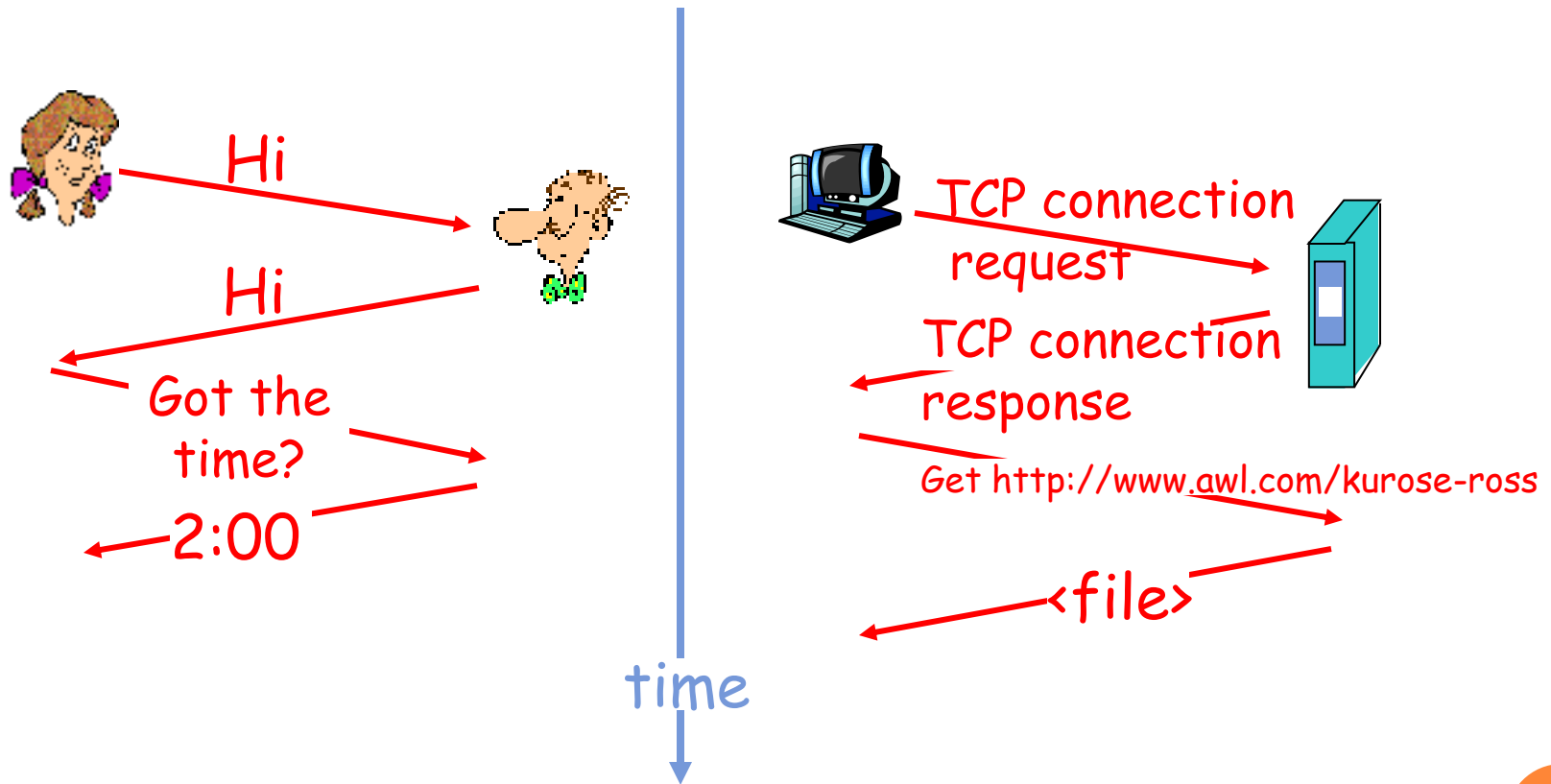
- machines rather than humans
- all communication activity in Internet governed by protocols

Protocols:

- protocols define format, order of messages sent and received among network entities, and actions taken on message transmission, receipt

WHAT'S A PROTOCOL?

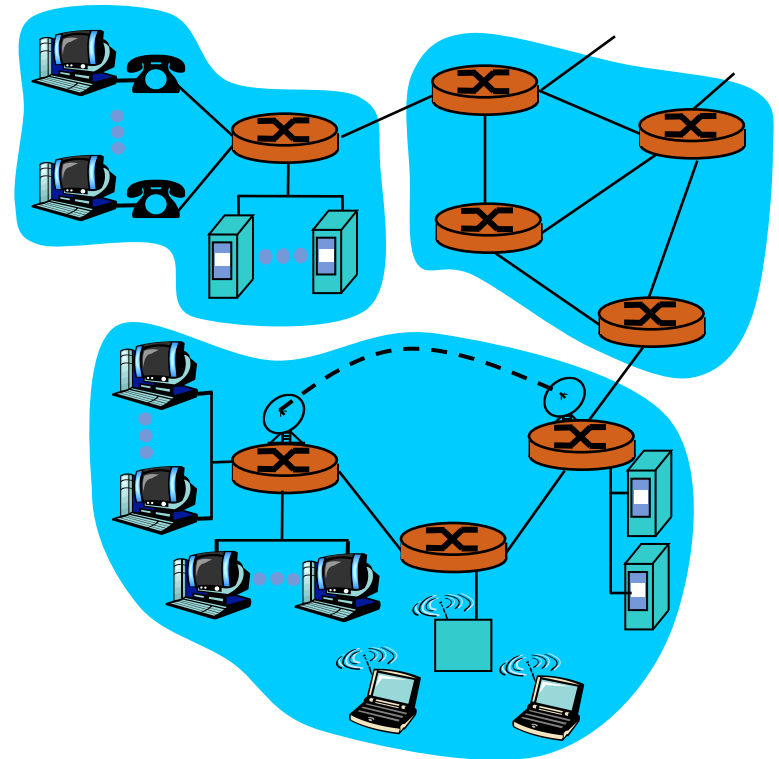
a human protocol and a computer network protocol:



Q: Other human protocols?

A CLOSER LOOK AT NETWORK STRUCTURE:

- network edge:
applications and hosts
- network core:
 - routers
 - network of networks
- access networks, physical media: communication links



THE NETWORK EDGE:

○ end systems (hosts):

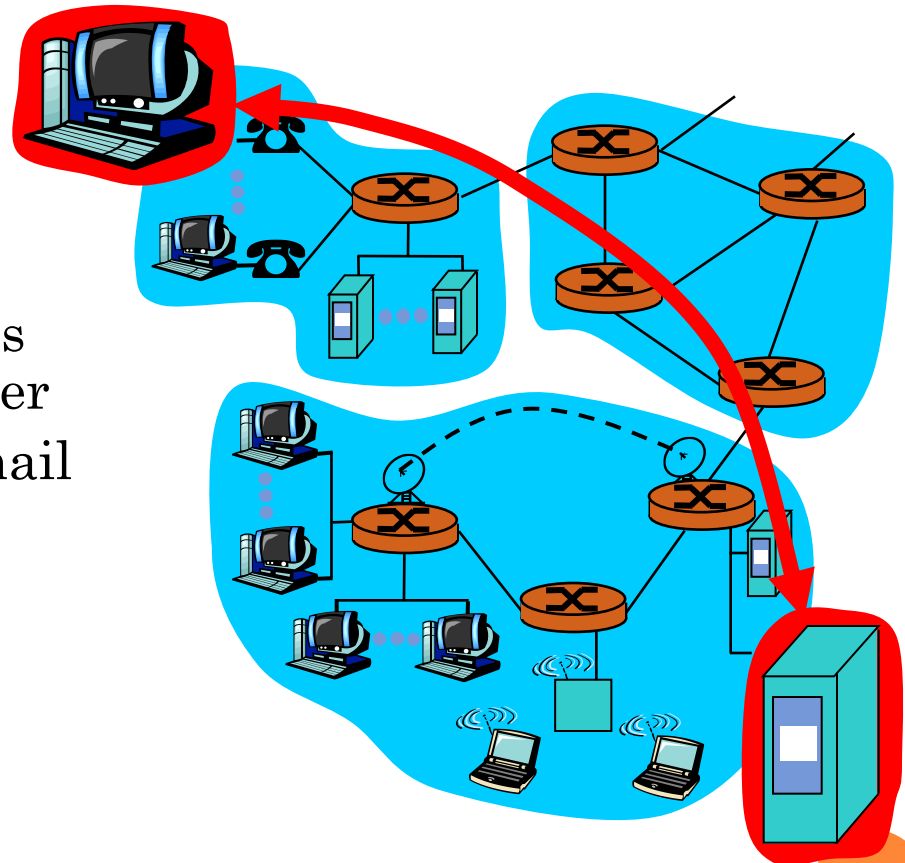
- run application programs
- e.g. Web, email
- at “edge of network”

○ client/server model

- client host requests, receives service from always-on server
- e.g. Web browser/server; email client/server

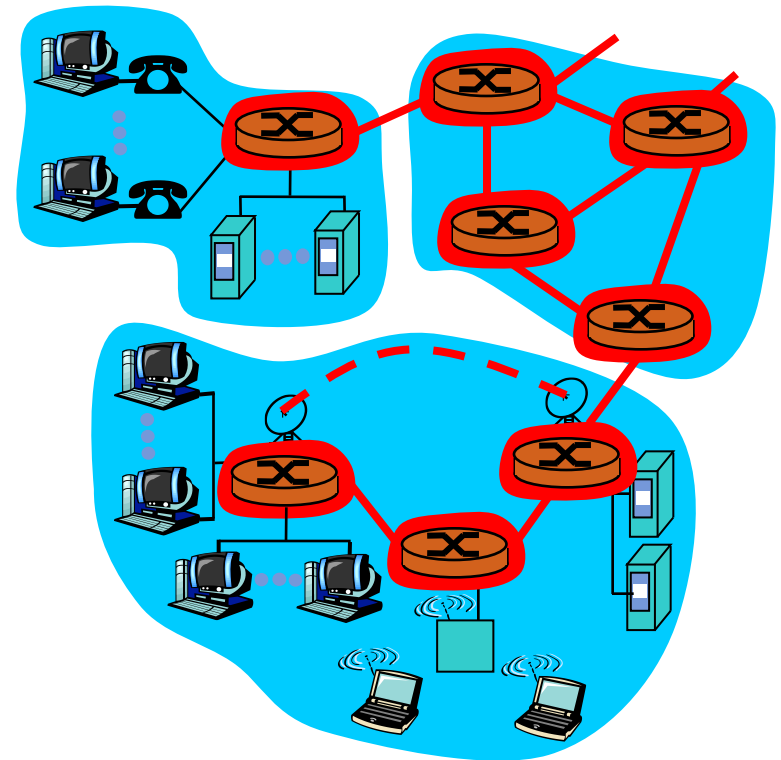
○ peer-peer model:

- minimal (or no) use of dedicated servers
- e.g. Skype, BitTorrent



THE NETWORK CORE

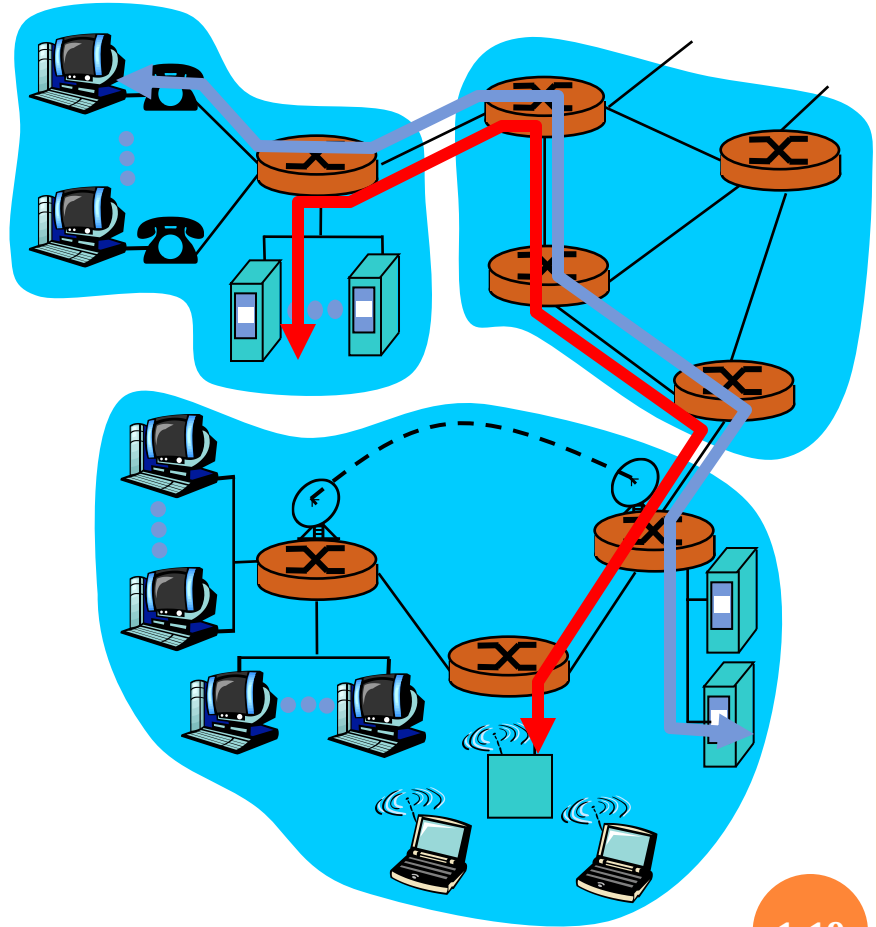
- mesh of interconnected routers
- *the fundamental question:*
how is data transferred through net?
 - **circuit switching:**
dedicated circuit per call:
telephone net
 - **packet-switching:** data
sent thru net in discrete
“chunks”



NETWORK CORE: CIRCUIT SWITCHING

End-end resources
reserved for “call”

- link bandwidth, switch capacity
- dedicated resources: no sharing
- circuit-like (guaranteed) performance
- call setup required



NETWORK CORE: CIRCUIT SWITCHING

Network resources (e.g., bandwidth) **divided into “pieces”**

- pieces allocated to calls
- resource piece *idle* if not used by owning call (*no sharing*)

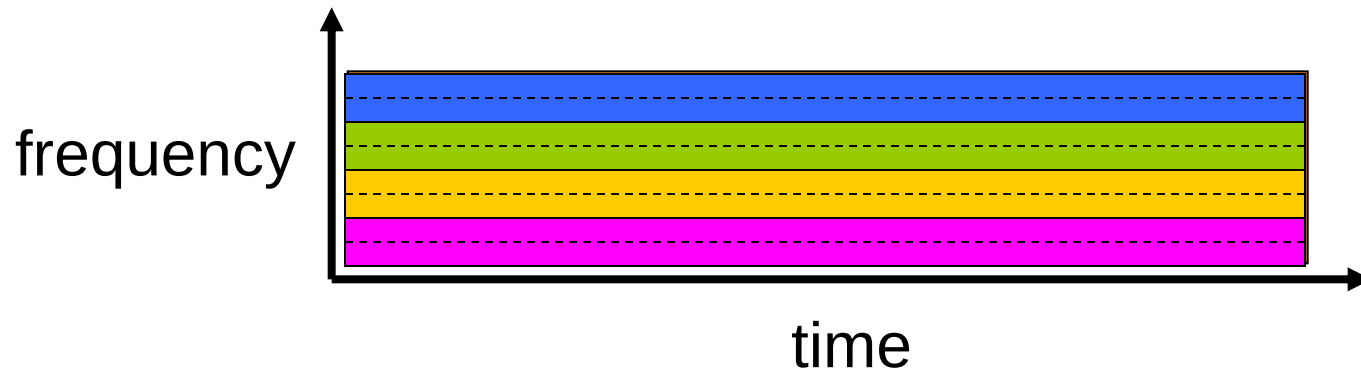
- dividing link bandwidth into “pieces”
 - ❖ frequency division
 - ❖ time division

CIRCUIT SWITCHING: FDM AND TDM

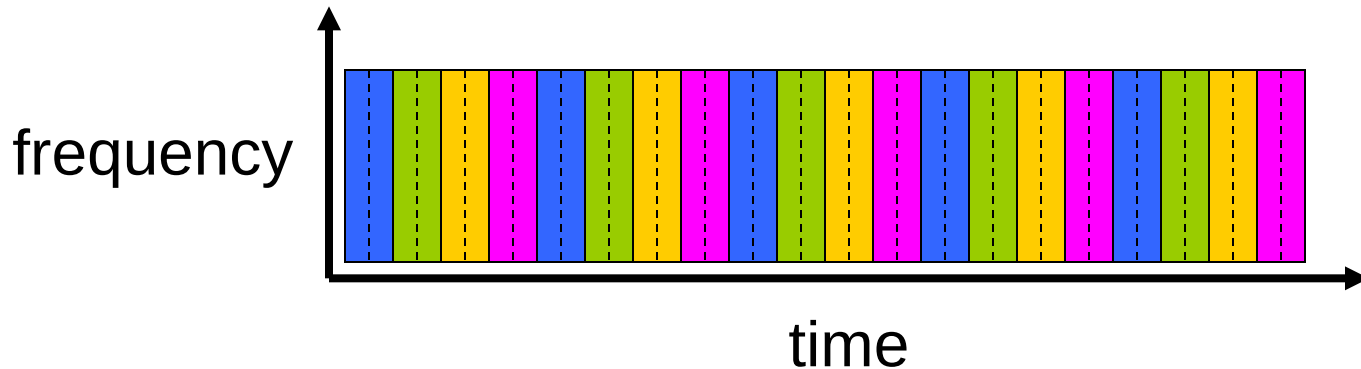
FDM

Example:

4 users



TDM



NUMERICAL EXAMPLE

- How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network?
 - All links are 1.536 Mbps
 - Each link uses TDM with 24 slots/sec
 - 500 msec to establish end-to-end circuit

Let's work it out!

NETWORK CORE: PACKET SWITCHING

each end-end data stream
divided into *packets*

- user A, B packets *share* network resources
- each packet uses full link bandwidth
- resources used *as needed*

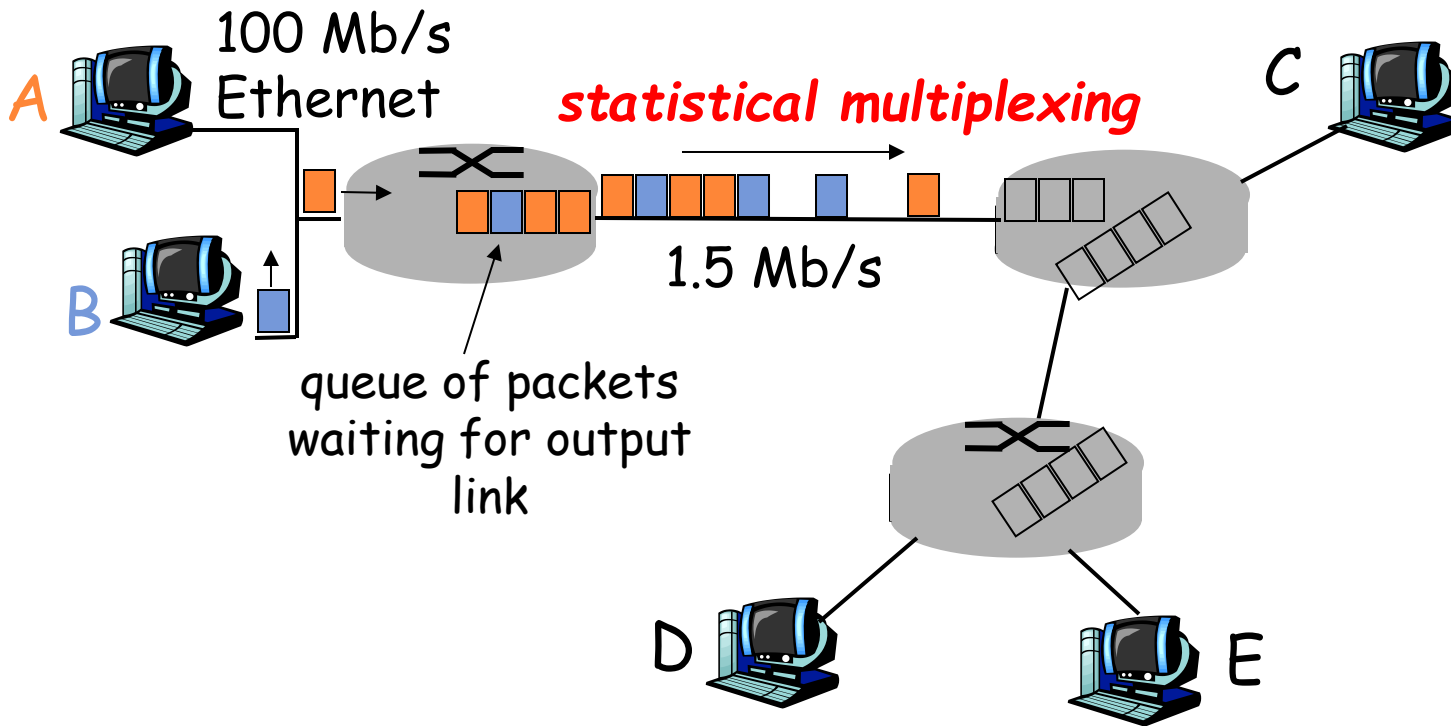
Bandwidth division into "pieces"
Dedicated allocation
Resource reservation



resource contention:

- ❑ aggregate resource demand can exceed amount available
- ❑ congestion: packets queue, wait for link use
- ❑ store and forward: packets move one hop at a time
 - ❖ Node receives complete packet before forwarding

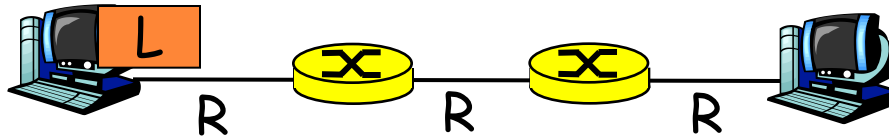
PACKET SWITCHING: STATISTICAL MULTIPLEXING



Sequence of A & B packets does not have fixed pattern, shared on demand □ *statistical multiplexing*.

TDM: each host gets same slot in revolving TDM frame.

PACKET-SWITCHING: STORE-AND-FORWARD



- Takes L/R seconds to transmit (push out) packet of L bits on to link or R bps
- Entire packet must arrive at router before it can be transmitted on next link: *store and forward*
- delay = $3L/R$ (assuming zero propagation delay)

Example:

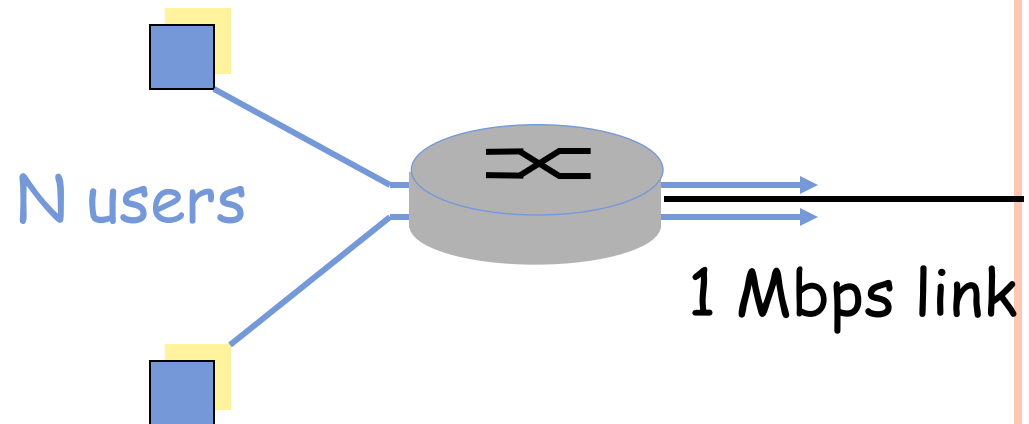
- $L = 7.5$ Mbits
- $R = 1.5$ Mbps
- delay = 15 sec

} more on delay shortly ...

PACKET SWITCHING VERSUS CIRCUIT SWITCHING

Packet switching allows more users to use network!

- 1 Mb/s link
- each user:
 - 100 kb/s when “active”
 - active 10% of time
- circuit-switching:
 - 10 users
- packet switching:
 - with 35 users, probability > 10 active less than .0004



Q: how did we get value 0.0004?

PACKET SWITCHING VERSUS CIRCUIT SWITCHING

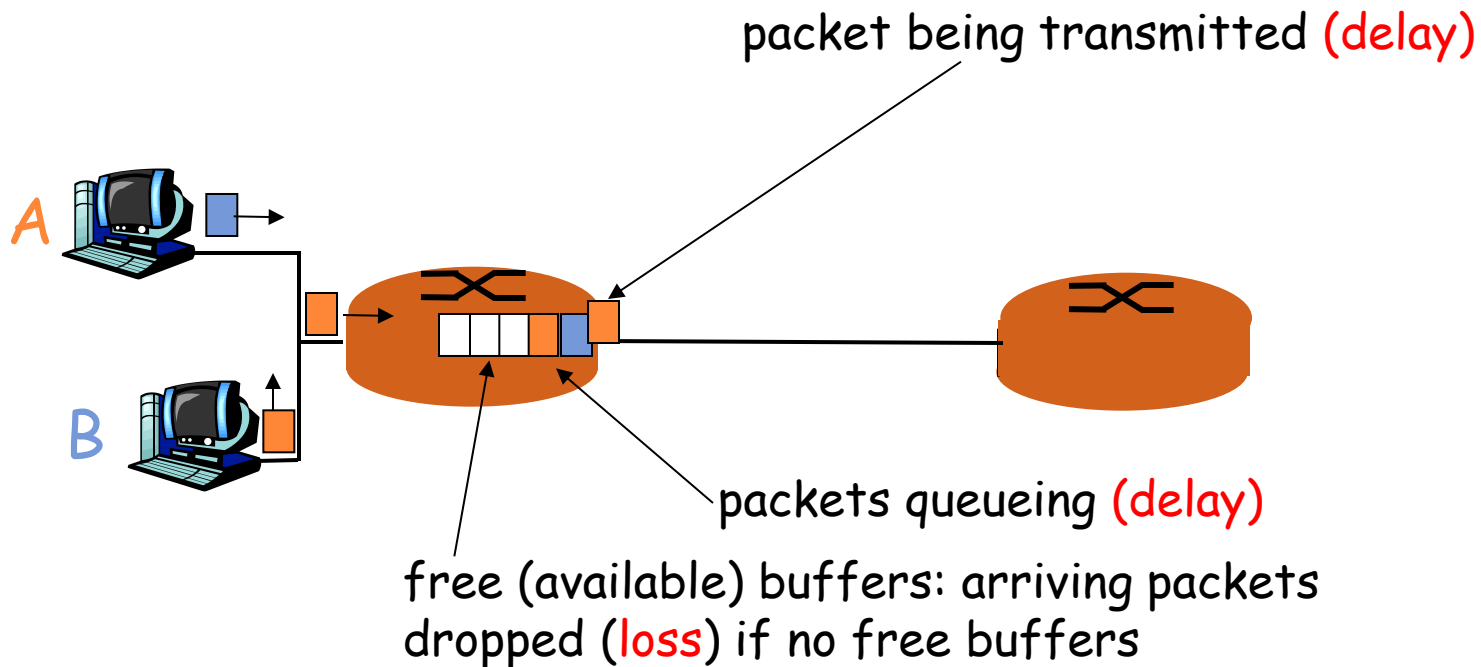
Is packet switching a “slam dunk winner?”

- Great for bursty data
 - resource sharing
 - simpler, no call setup
- **Excessive congestion:** packet delay and loss
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior?**
 - bandwidth guarantees needed for audio/video apps
 - still an unsolved problem

HOW DO LOSS AND DELAY OCCUR?

packets *queue* in router buffers

- packet arrival rate to link exceeds output link capacity
- packets queue, wait for turn



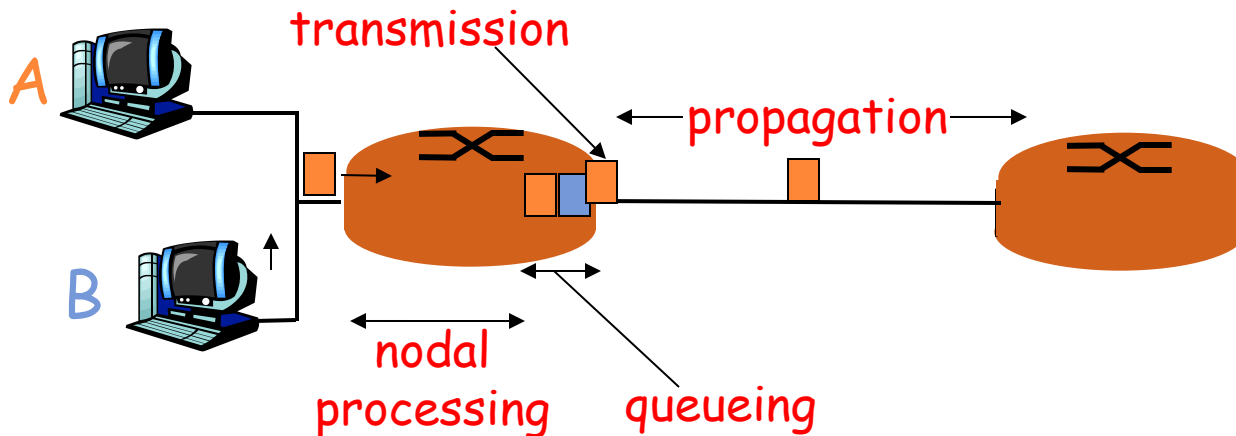
FOUR SOURCES OF PACKET DELAY

1. nodal processing:

- check bit errors
- determine output link

2. queueing

- ❖ time waiting at output link for transmission
- ❖ depends on congestion level of router



DELAY IN PACKET-SWITCHED NETWORKS

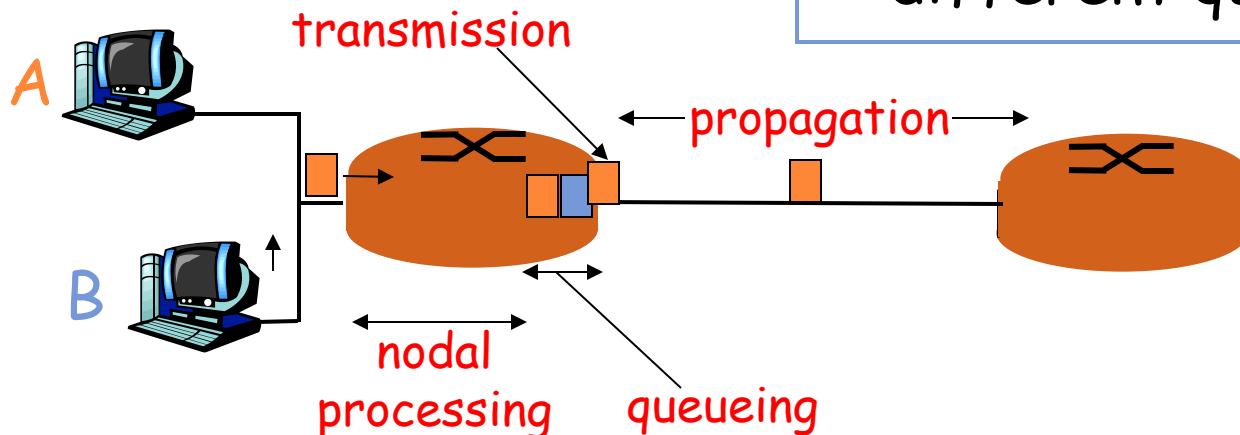
3. Transmission delay:

- R = link bandwidth (bps)
- L = packet length (bits)
- time to send bits into link = L/R

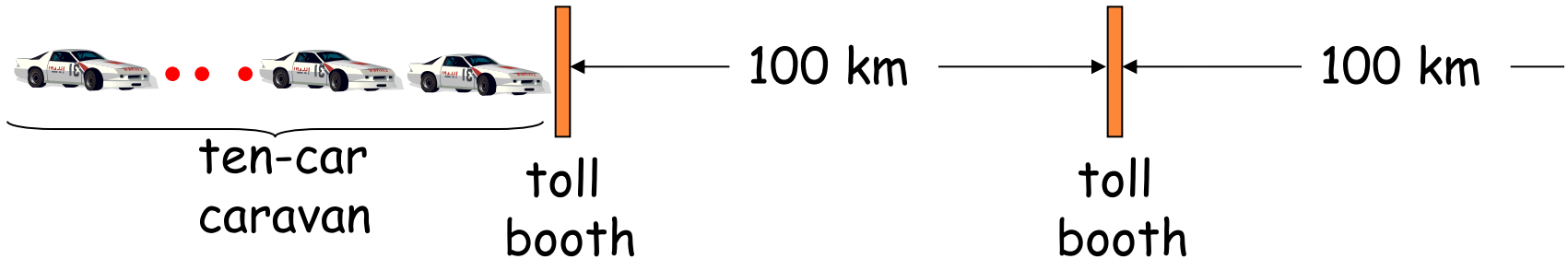
4. Propagation delay:

- d = length of physical link
- s = propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- propagation delay = d/s

Note: s and R are very different quantities!

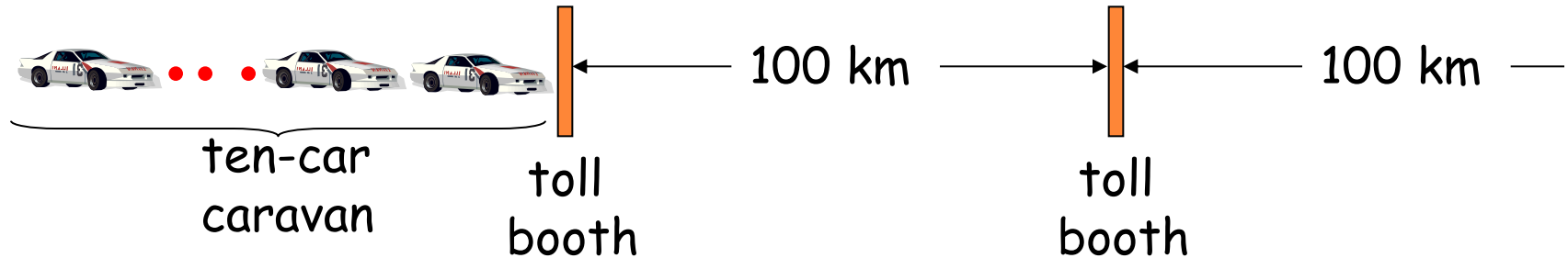


CARAVAN ANALOGY



- Cars “propagate” at 100 km/hr
- Toll booth takes 12 sec to service a car (transmission time)
- car~bit; caravan ~ packet
- Q: How long until caravan is lined up before 2nd toll booth?
- Time to “push” entire caravan through toll booth onto highway = $12 \times 10 = 120$ sec
- Time for last car to propagate from 1st to 2nd toll booth:
 $100\text{km} / (100\text{km/hr}) = 1$ hr
- A: 62 minutes

CARAVAN ANALOGY (MORE)



- Cars now “propagate” at 1000 km/hr
- Toll booth now takes 1 min to service a car
- Q: Will cars arrive to 2nd booth before all cars serviced at 1st booth?
- Yes! After 7 min, 1st car at 2nd booth and 3 cars still at 1st booth.
- 1st bit of packet can arrive at 2nd router before packet is fully transmitted at 1st router!
 - See Ethernet applet at AWL Web site

NODAL DELAY

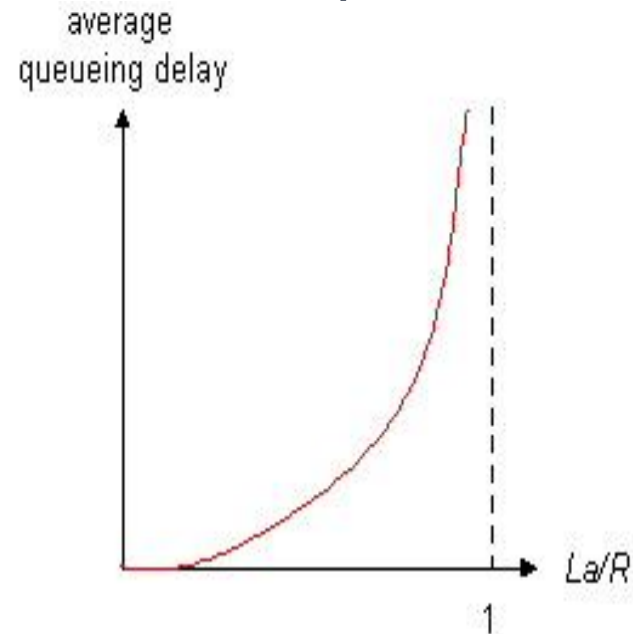
$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- d_{proc} = processing delay
 - typically a few microsecs or less
- d_{queue} = queuing delay
 - depends on congestion
- d_{trans} = transmission delay
 - $= L/R$, significant for low-speed links
- d_{prop} = propagation delay
 - a few microsecs to hundreds of msecs

QUEUEING DELAY (REVISITED)

- R =link bandwidth (bps)
- L =packet length (bits)
- a =average packet arrival rate

traffic intensity = La/R



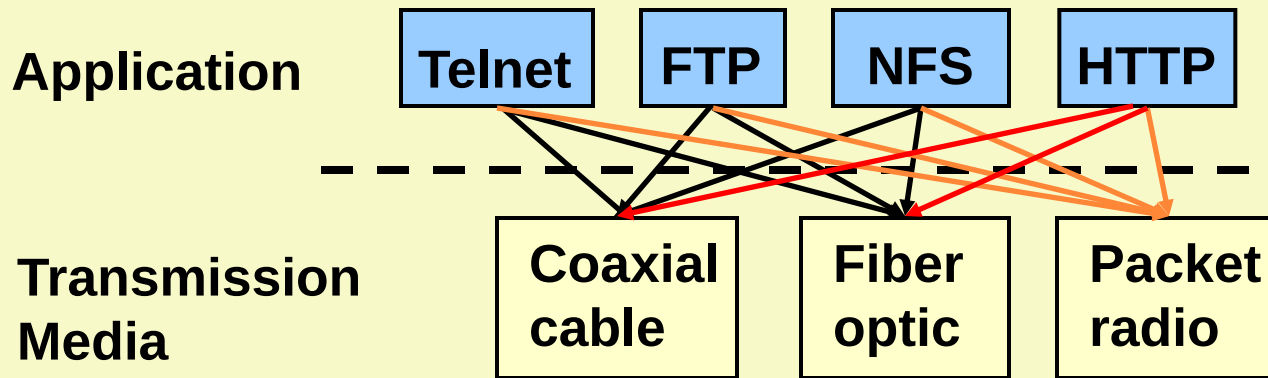
- $La/R \sim 0$: average queueing delay small
- $La/R \rightarrow 1$: delays become large
- $La/R > 1$: more "work" arriving than can be serviced, average delay infinite!

SIGNIFICANCE OF LAYERED NETWORK ARCHITECTURE

- ❑ Why layering?
 - ❑ Dealing with complex systems:
 - explicit structure allows identification, relationship of complex system's pieces
 - layered **reference model** for discussion
 - ❑ Modularization eases maintenance, updating of system
 - change of implementation of layer's service transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system

WHY LAYERING?

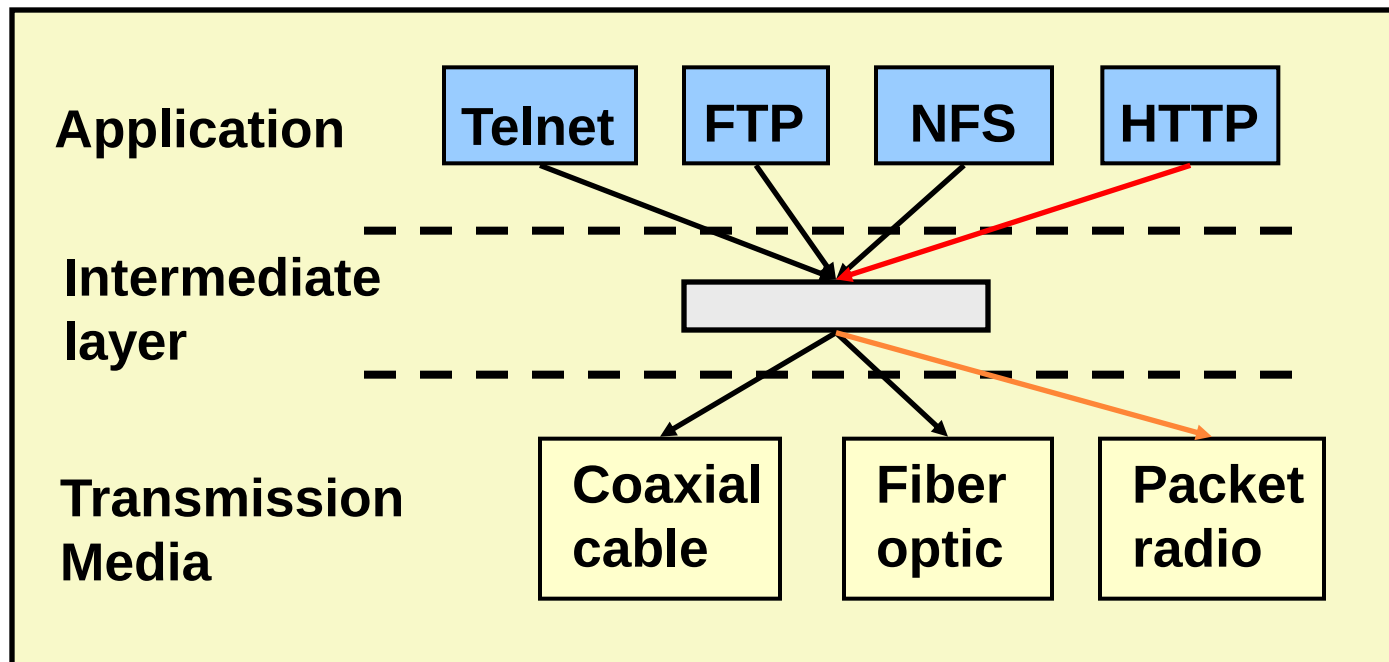
(FTP – File Transfer Protocol, NFS – Network File Transfer, HTTP – World Wide Web protocol)



- No layering: each new application has to be *re-implemented* for every network technology!

WHY LAYERING?

- Solution: introduce an intermediate layer that provides a unique abstraction for various network technologies



LAYERING

- **Layering**:-A technique to organize a network system into a **succession** of logically distinct entities, such that the service provided by one entity is **solely** based on the service provided by the previous (lower level) entity
- **Advantages**
 - Modularity – protocols easier to manage and maintain
 - Abstract functionality –lower layers can be changed **without** affecting the upper layers
 - Reuse – upper layers can reuse the functionality provided by lower layers
- **Disadvantages**
 - Information hiding – inefficient implementations

STANDARDIZED PROTOCOL ARCHITECTURES

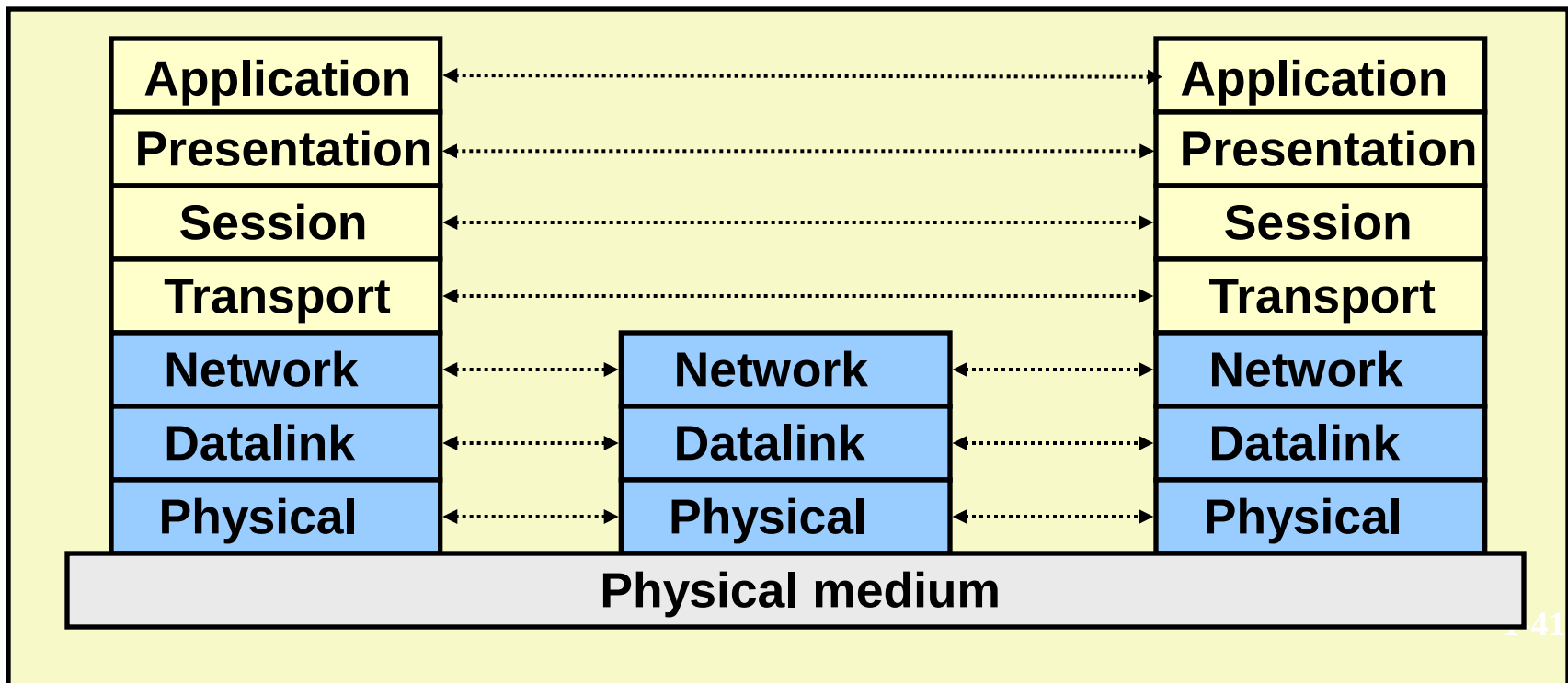
- Required for devices to communicate
- Vendors have more marketable products
- Customers can insist on standards based equipment
- Two standards:
 - OSI Reference model
 - Never lived up to early promises
 - TCP/IP protocol suite
 - Most widely used

OSI - THE MODEL

- Open Systems Interconnection (OSI) is a layered model
- Developed by the International Organization for Standardization (ISO)
- A theoretical system delivered too late! and has Seven layers
- Each layer performs a subset of the required communication functions
- Each layer relies on the next lower layer to perform more primitive functions
- Each layer provides services to the next higher layer
- Changes in one layer should not require changes in other layers

ISO OSI REFERENCE MODEL

- Seven layers
 - Lower three layers are peer-to-peer
 - Next four layers are end-to-end



OSI LAYERS (1)

○ Physical

- Physical interface between devices
 - Mechanical (joins 1 or more signal conductor, circuits)
 - Electrical (Representation of bits and bit rate)
 - Functional (function performed by individual circuits)
 - Procedural (sequence of events)
- Modulation and Demodulation
- Raw bit stream
- Modem: broadly used to refer to any module that performs the function above

OSI LAYERS(2)- DATA LINK LAYER

- Data Link Layer
 - Means of activating, maintaining and deactivating a reliable link
 - Framing (Header and Trailer)
 - Physical addressing
 - Error detection
 - Flow control
 - Error Control
 - Higher layers may assume error free transmission

OSI LAYERS (4)

○ Network Layer

- Transport of information
- Virtual circuit service (connection oriented)
- Packet switching or datagram service (connection less)
- Logical addressing (IP address)
- Routing
- Congestion control (flow of packet into the network)
- Both connection-less and connection-oriented

OSI LAYERS(5)

○ Transport

- Exchange of data between end systems
- Error free
- In sequence (segment)
- No losses
- No duplicates
- Quality of service (Throughput, transit delay, error rate)

OSI LAYERS (6)

- Session
 - Control of dialogues between applications/Dialogue discipline
 - Grouping
 - Synchronization/check points
 - Recovery
- Presentation
 - Data formats
 - Architecture specific (Big-endian or Little-endian)
 - Provide conversion from one encoding schema to another encoding schema
 - Data compression
 - Encryption
- Application
 - Means for applications to access OSI environment
 - E mail, web browsers,

TCP/IP PROTOCOL ARCHITECTURE

- Developed by the US Defense Advanced Research Project Agency (DARPA) for its packet switched network Advanced Research Projects Agency Network (ARPANET)
- Used by the global Internet
- No official model but a working one.
 - Application layer
 - Host to host or transport layer
 - Internet layer
 - Data link / Network access layer
 - Physical layer

PHYSICAL LAYER

- Similar to OSI model physical layer
- Physical interface between data transmission device (e.g. computer) and transmission medium or network
- Characteristics of transmission medium
- Signal levels
- Data rates

NETWORK ACCESS/DATA LINK LAYER(1)

- Data Link Control(Logical Link Control) Sub Layer
 - Means of activating, maintaining and deactivating a reliable link
 - Framing (Header)
 - Error detection and control
- Media Access Control (MAC) sub layer
 - Allocation of channel
 - Physical addressing (MAC address)
 - Higher layers do not need to know about underlying technology
 - Virtual point-point links between pair of stations

IP LAYER

- Exchange of data between end system and network
- Systems may be attached to different networks
- Datagram
- Implemented in end systems and routers
- Invoking services like priority
- Connection less service
- Fragmentation
- Routing and IP addresses
- ARP and RARP

TRANSPORT LAYER (TCP/UDP)

- Connection-less and connection oriented service
- Reliable delivery of data
- Segments
- Congestion control
- Ordering of delivery(Reassembling)
- ❑ Architecture specific (Big-endian or Little-endian)
- Addressing (Port no. or SAP)
- UDP (TFTP, NFS, DNS)
- TCP (SMTP, HTTP, FTP)
- Error Control

APPLICATION LAYER

- Support for user applications
- File transfer
- User applications
- Reliable data transfer for UDP users
- Network management
- E.g. http, SMTP, FTP, etc.

REFERENCES

- William Stallings, “Data and Computer Communications”, Seventh Edition, PHI 2004.
- Andrew S. Tanenbaum, ”Computer Networks” 4th Edition PHI
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- James F. Kurose, Keith W. Ross, “Computer Networking: A Top-Down Approach Featuring the Internet”, 3rd Edition , Pearson Education 2009.
- https://gaia.cs.umass.edu/kurose_ross/ppt.htm
- PPT available for the respective books

THANK YOU

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